First Problem Sheet
on
Linear Algebra

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Topic: Elementary Group Theory

"The theory of groups is the most universal language of symmetry, and symmetry, as wide or as narrow as you may define its meaning, is one idea by which man through the ages has tried to comprehend and create order, beauty, and perfection."
-Hermann Weyl

Problem 1

Show that

i) The identity element of a group is unique.
ii) The inverse of a given element of a group is unique.
iii) $(gh)^{-1} = h^{-1}g^{-1}$ for any g, h elements of a group.

Problem 2

Let $GL(\mathcal{V})$ denote the set of all invertible operators of a linear space $\mathcal{V}$. In particular, we write $GL(n, \mathbb{C})$ when $\mathcal{V} = \mathbb{C}^n$ with similar notation for $\mathbb{R}$. Show that $GL(\mathcal{V})$ forms a group with respect to the composition of operators, called the **general linear group** of $\mathcal{V}$. Let $SL(n, \mathbb{R})$ be the subset of $GL(n, \mathbb{R})$ consisting of elements having unit determinant and hence show that $SL(n, \mathbb{R})$ is a proper subgroup of $GL(n, \mathbb{R})$, called the **special linear group**.

Problem 3

Let $x, y \in \mathbb{R}^n$, and define an inner product on $\mathbb{R}^n$ by

$$\langle x, y\rangle = -x_1y_1 - ... - x_py_p + x_{p+1}y_{p+1} + ... + x_ny_n.$$

Denote the subset of $GL(n, \mathbb{R})$ that leaves this inner product invariant by $O(p, n-p)$. Show that $O(n, n-p)$ is a subgroup of $GL(n, \mathbb{R})$. The set of linear transformations among $O(n, n-p)$ that have determinant 1 is denoted by $SO(n, n-p)$. The special case of $p = 0$ gives us the orthogonal and special orthogonal groups. When $n = 4$ and $n = 3$, we get the inner product of the special theory of relativity, and $O(3, 1)$, the set of Lorentz transformations, is called the **Lorentz group**. If one adds translations of $\mathbb{R}^4$ to $O(3, 1)$, one obtains the **Poincaré group**, $P(3, 1)$.

Problem 4

Show that homomorphisms sends identity to identity and inverse to inverse.